

MSDS 7333 - Case Study 2: Analysis and Modeling of Diabetic Data

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Contents

Chapter 1

Introduction

1.1 Background

Diabetes represents a significant global health challenge, affecting millions of individuals worldwide and contributing substantially to healthcare expenditure. The condition is characterized by elevated blood glucose levels resulting from defects in insulin production, insulin action, or both. The complications associated with diabetes include cardiovascular disease, nephropathy, retinopathy, and neuropathy, significantly impacting patient quality of life and mortality rates.

Hospital readmissions among diabetic patients constitute a significant burden on healthcare systems, with implications for both quality of care and resource allocation. Readmissions within 30 days of discharge are particularly concerning as they may indicate inadequate initial treatment, premature discharge, or insufficient follow-up care. Understanding the factors associated with readmission patterns is essential for developing effective intervention strategies to improve patient outcomes and reduce healthcare costs.

1.2 Dataset Description

This analysis utilizes a comprehensive dataset containing records of diabetic patient encounters from 1999-2008 at 130 U.S. hospitals. The dataset comprises 101,766 observations across 50 variables, encompassing patient demographics, admission details, diagnoses, laboratory tests, and medication regimens. The target variable, "readmitted," categorizes patient outcomes into three classes: readmitted within 30 days (" <30 "), readmitted after 30 days (" >30 "), or not readmitted ("NO").

Key features in the dataset include:

- Demographic information ('age', 'race', 'gender', 'weight')
- Admission type, source, and discharge disposition ('admission_{type}*d*', 'discharge_{disposition}*d*', 'admission_{source}*d*')
- Primary, secondary, and tertiary diagnoses ('diag₁', 'diag₂', 'diag₃')
- Laboratory tests ('A1Cresults', 'max_{glucose}*erum*')
- Medication changes during hospitalization ('num_{medications}*s*')
- Diabetes medications (e.g., insulin, metformin, glyburide)

- Length of hospital stay ('time_in_{hospital}')
- Number of previous outpatient, inpatient, and emergency visits

1.3 Research Objectives

The primary objectives of this analysis are to:

1. Identify patterns and relationships within the diabetic patient data through comprehensive exploratory data analysis
2. There exists several missing values in the dataset, we need to handle and impute them appropriately
3. Develop a predictive model using logistic regression with k-fold cross-validation to estimate readmission probabilities
4. Evaluate the performance of the model and identify the most influential features affecting readmission outcomes
5. Determine the key factors associated with hospital readmission among diabetic patients

Through this investigation, we aim to contribute insights that may inform clinical decision-making and intervention strategies to reduce readmission rates among diabetic patients.

Chapter 2

Exploratory Data Analysis

2.1 Data Overview and Preprocessing

The exploratory data analysis began with an examination of the dataset structure and composition. The diabetic dataset comprises 101,766 patient encounters with 50 variables.

The distribution of the target variable "readmitted" reveals that 54,864 (53.9%) patients were not readmitted, 35,545 (34.9%) were readmitted after 30 days, and 11,357 (11.2%) were readmitted within 30 days (Figure 2.1). The target variable is imbalanced, which is a common issue in healthcare datasets.

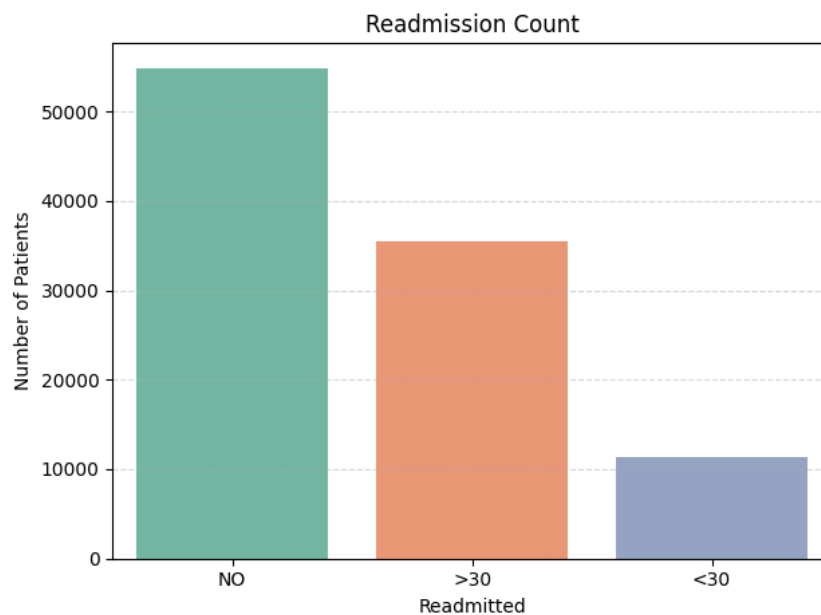


Figure 2.1: Distribution of patient readmission categories

Initial preprocessing steps included:

- Removal of identifier variables (encounter_id and patient_nbr)
- Replacement of "?" placeholders with NaN values for proper missing value handling

2.2 Missing Value Analysis and Imputation

A critical aspect of the preprocessing phase involved identifying and addressing missing values within the dataset. Nine variables exhibited missing values, with "weight" showing the highest proportion at 96.9%, followed by "max_glu_serum" and "A1Cresult" at 94.75% and 83.28% respectively.

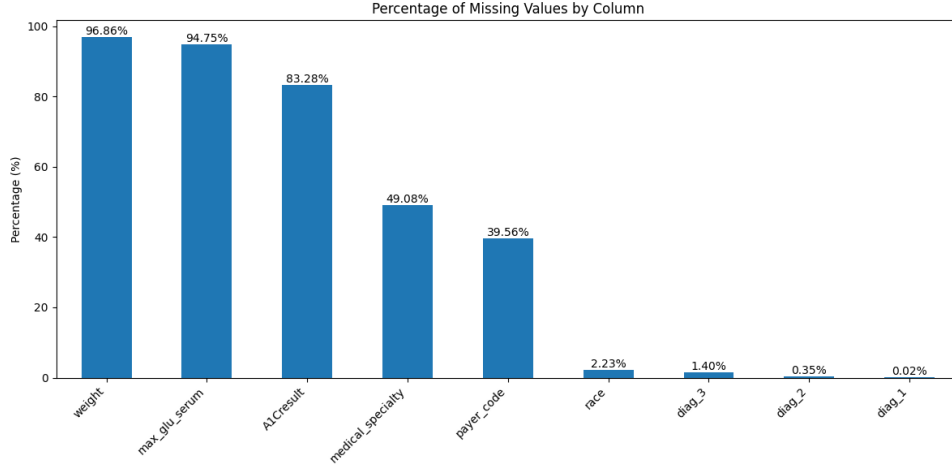


Figure 2.2: Percentage of missing values by variable

The following imputation strategies were implemented:

- Although "weight" is very important factor to identify the diabete condition, however, due to its extremely high percentage of missing values (96.9%) we dropped this variable.
- Despite its high percentage of missing values, due to small amount of categorical objects, the "max_glu_serum", "A1Cresult", and "medical_specialty", were imputed with the value "None"
- Similar to the above categorical variables, the "payer_code" and "race" variables were imputed with the value "None"
- For diagnosis variables: "diag_1", "diag_2", "diag_3", missing values were imputed with the value "None". Categories with less than 0.5% occurrence were merged into a new "Rest" category to reduce dimensionality and prevent overfitting. Due to its importance, the "diag_1" variable was considered more important than "diag_2", and "diag_2" was considered more important than "diag_3". If a patient has multiple diagnoses, the most important one was selected and put into the "diag" variable. The "Rest" category was created to handle the rare cases and thus has less importance ranking (after diag_3). The "None" category was created to handle the missing values and has the least importance (after "Rest"). This approach was used to handle the overfitting issue and reduce the dimensionality of the dataset.
- For "race", the missing values were imputed with "None", although it could be aligned with "Caucasian" or "AfricanAmerican" categories, due to high median values.
- For "age", the missing values were imputed with the median of the variable. ("age_mid")

2.3 Categorical Variable Analysis

The dataset contains numerous categorical variables representing patient demographics, diagnoses, medications, and hospital admission details. Categorical variables were categorized into two groups based on their distribution patterns and relationship with the target variable.

2.3.1 Key Categorical Variables

Based on distribution analysis and correlation with the target variable, the following categorical variables were identified as significant:

- Demographics: race, gender, payer_code (Figure 2.3)
- Medical details: max_glu_serum, A1Cresult, medical_specialty (Figure 2.4)
- Medications: metformin, glimepiride, glipizide, glyburide, pioglitazone, rosiglitazone (Figure 2.5)
- Treatment process: insulin_change, diabetesMed (Figure 2.6)
- Diagnosis: diag (Figure 2.7)
- Hospital factors: admission_type_id, discharge_disposition_id, admission_source_id (Figure 2.8)

Notable findings include:

- Race: Over 70% of patients are Caucasian, with distinct readmission patterns across racial groups
- Admission type: Emergency intake (type 1) accounts for more than 50% of admissions
- Discharge disposition: Approximately 60% of patients are discharged to home
- Admission source: Nearly 60% of admissions originate from emergency rooms, followed by 30% from physician referrals

2.3.2 Excluded Categorical Variables

Several medication variables were excluded from further analysis due to extreme skewness that could lead to overfitting:

- repaglinide, nateglinide, chlorpropamide, acetohexamide, tolbutamide
- acarbose, miglitol, troglitazone, tolazamide, examide
- citoglipton, glyburide_metoformin, glipizide_metoformin
- glimepiride_glitazone, troglitazone, metformin_pioglitazone

2.4 Numerical Variable Analysis

The numerical variables in the dataset include:

- `time_in_hospital`: Length of hospital stay
- `num_lab_procedures`: Number of laboratory procedures
- `num_procedures`: Number of procedures other than laboratory
- `num_medications`: Number of distinct medications
- `number_outpatient`: Number of outpatient visits
- `number_emergency`: Number of emergency visits
- `number_inpatient`: Number of inpatient visits
- `number_diagnoses`: Number of diagnoses
- `age_mid`: Midpoint of age range

2.4.1 Correlation Analysis

Correlation analysis was conducted to identify potential relationships between numerical variables. The correlation matrix did not reveal high collinearity between the variables, with most correlation coefficients below 0.4.

Variance Inflation Factor (VIF) analysis was also performed to detect multicollinearity. Although some potential collinearity was observed between "number_diagnoses" and "age", the VIF values remained within acceptable ranges, and no variables were removed based on this analysis.

2.4.2 Distribution Analysis

Distribution analysis of numerical variables revealed:

- Right-skewed distributions for `time_in_hospital`
- Left-skewed distribution for `age`
- Extreme right-skew for `number_outpatient`, `number_emergency`, and `number_inpatient`, with most values being zero

Due to the extreme skew in outpatient, emergency, and inpatient visit counts, these variables were transformed into categorical variables:

- `cat_outpatient`: 0 for no visits, 1 for one or more visits
- `cat_emergency`: 0 for no visits, 1 for one or more visits
- `cat_inpatient`: 0 for no visits, 1 for one visit, 2 for more than one visit

2.4.3 Relationship with Target Variable

Boxplots were used to visualize the relationship between numerical variables and the readmission status. The analysis showed comparable distributions across the three readmission categories, indicating that standardization of numerical variables would be appropriate before integration with categorical features.

2.5 Feature Engineering and Preparation for Modeling

After comprehensive exploratory analysis, the following preprocessing steps were implemented to prepare the data for modeling:

1. Merging of the transformed categorical variables (cat_outpatient, cat_emergency, cat_inpatient) with the selected categorical features
2. One-hot encoding of categorical variables, resulting in 193 binary features
3. Standardization of remaining numerical variables
4. Integration of encoded categorical features with standardized numerical features to form the final feature matrix

This preprocessing pipeline transformed the original dataset with 50 variables into a modeling-ready matrix while preserving the information content and addressing the identified data quality issues.

2.6 Feature Importance Analysis

To gain deeper insights into the predictive power of individual features, a Random Forest classifier was trained on the preprocessed data. Feature importance scores were extracted to identify the most influential variables in predicting hospital readmission.

The analysis revealed several key insights:

- Hospital-related factors such as discharge_disposition_id and time_in_hospital emerged as the most important predictors of readmission
- Patient demographics including age and race also showed significant influence on readmission risk
- Clinical factors like number_diagnoses, number_inpatient visits, and number_emergency visits were strong predictors
- Among medication variables, insulin treatment demonstrated the highest importance, supporting existing literature on the relationship between insulin therapy and readmission risk

These findings provide valuable insights for healthcare providers to identify high-risk patients and develop targeted interventions to reduce readmission rates. The prominence of hospital-related factors suggests that improvements in discharge planning and post-discharge follow-up could be effective strategies for reducing readmissions.

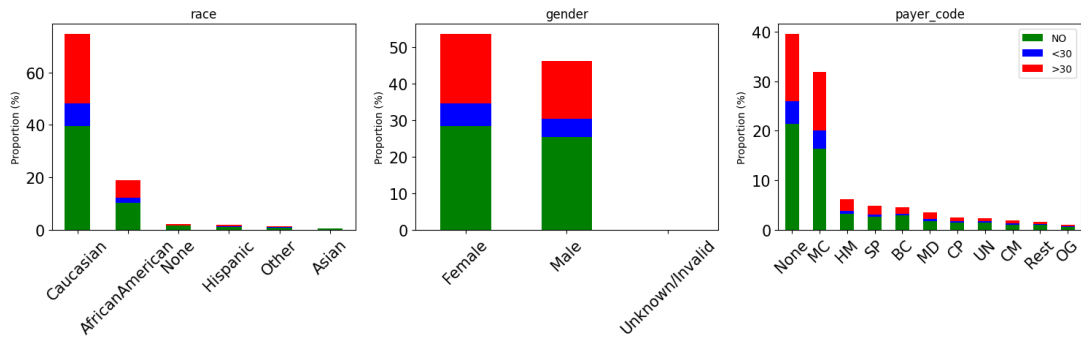


Figure 2.3: Distribution of readmission status across Demographics

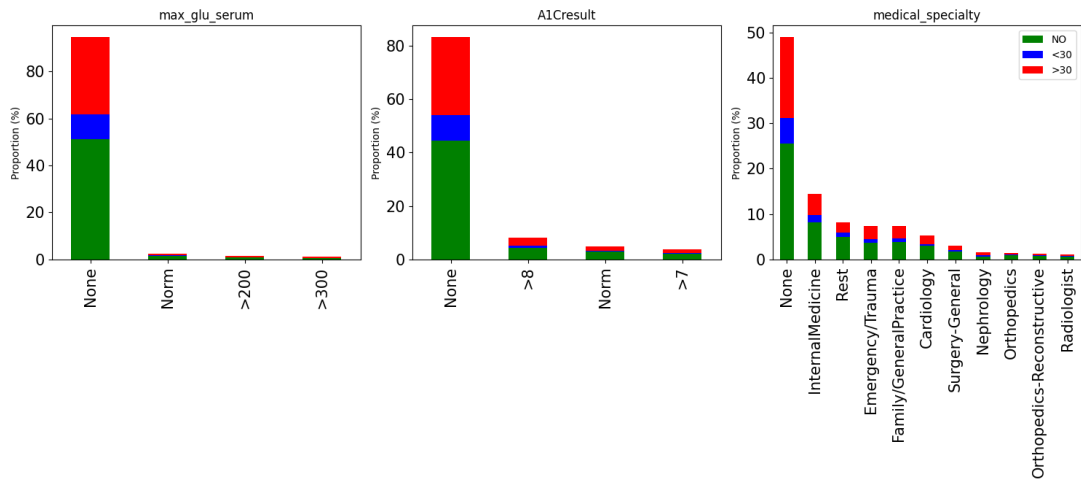


Figure 2.4: Distribution of readmission status across Medical details

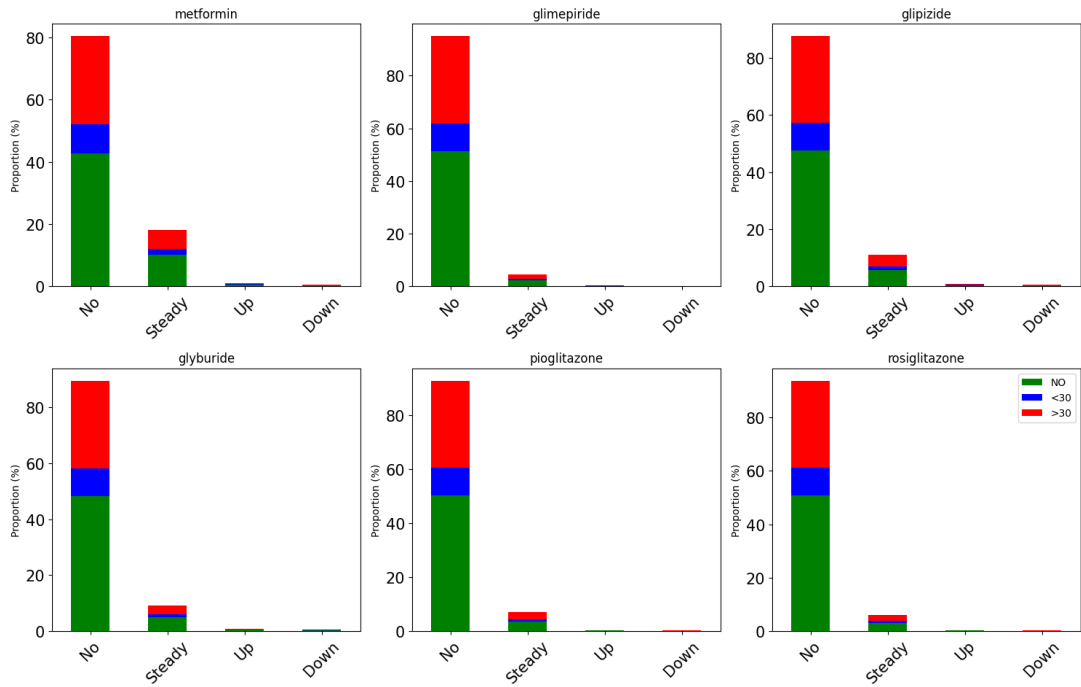
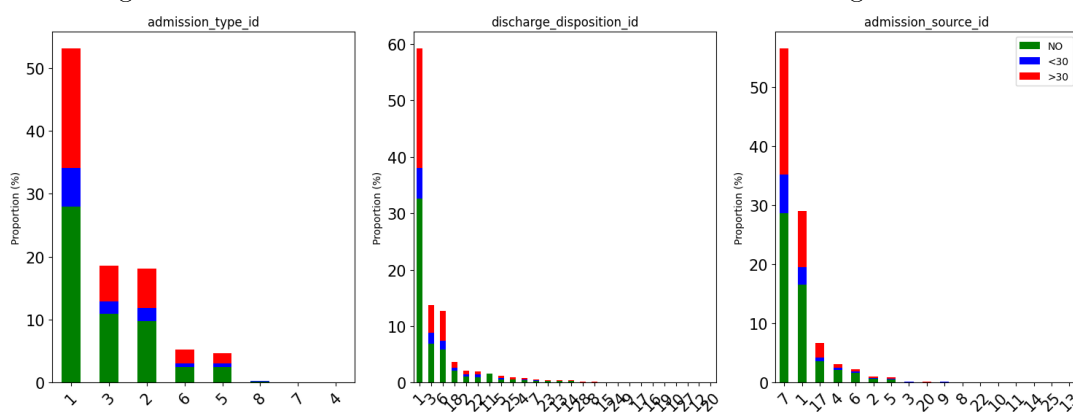
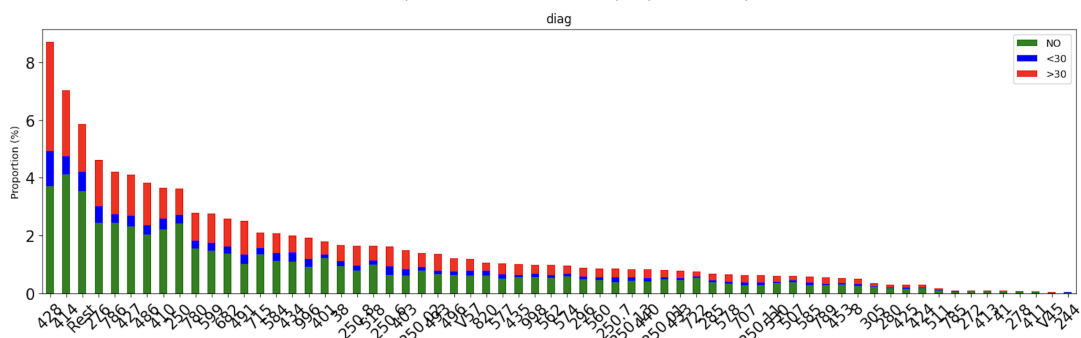
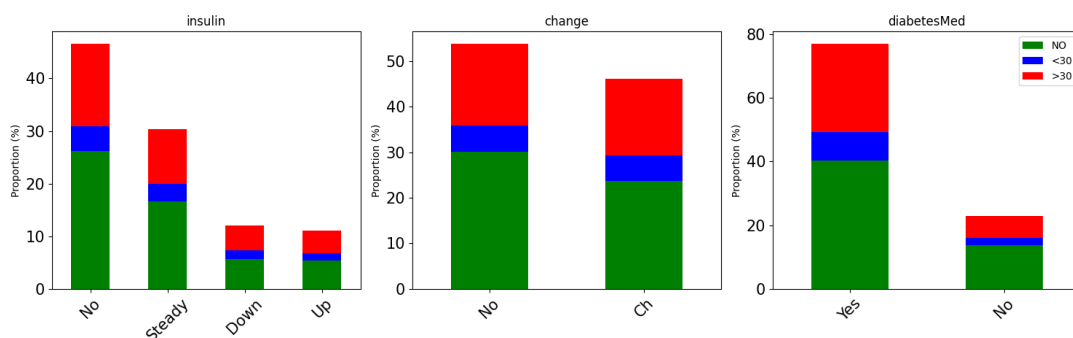


Figure 2.5: Distribution of readmission status across Medications



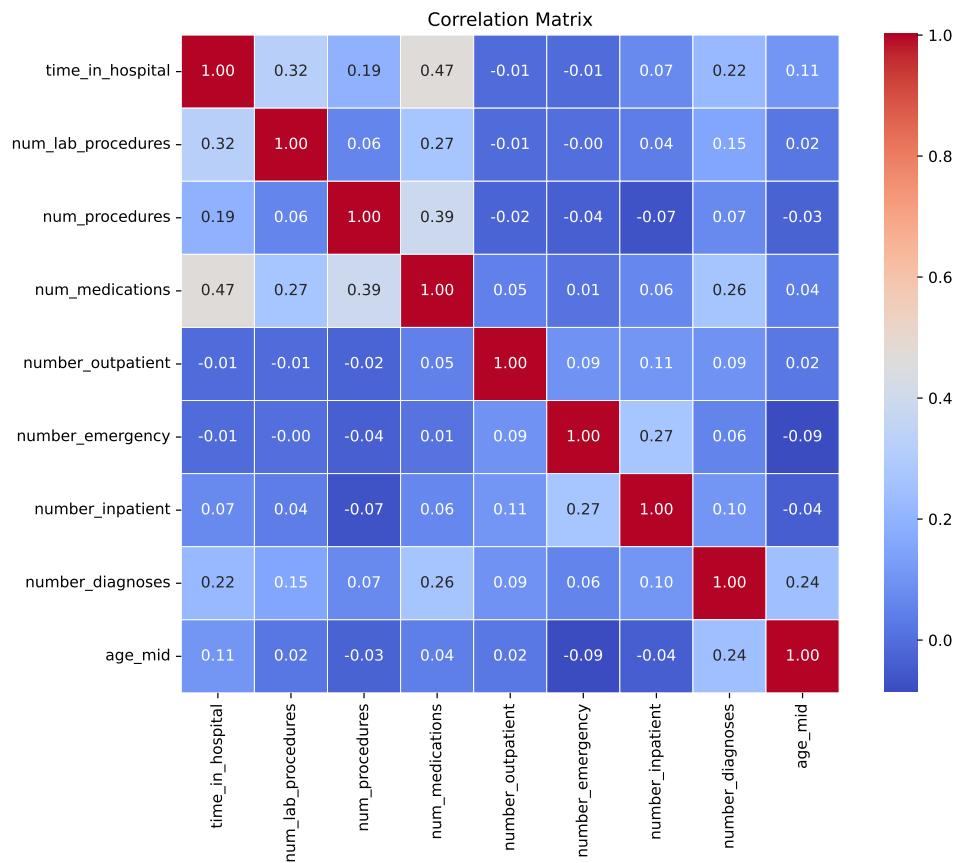


Figure 2.9: Correlation matrix of numerical variables

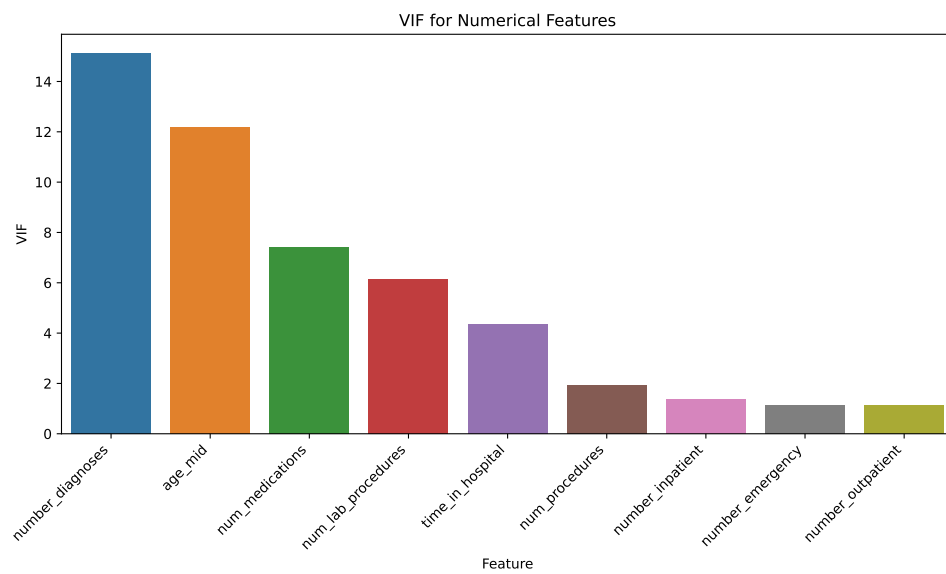


Figure 2.10: Variance Inflation Factor for numerical features

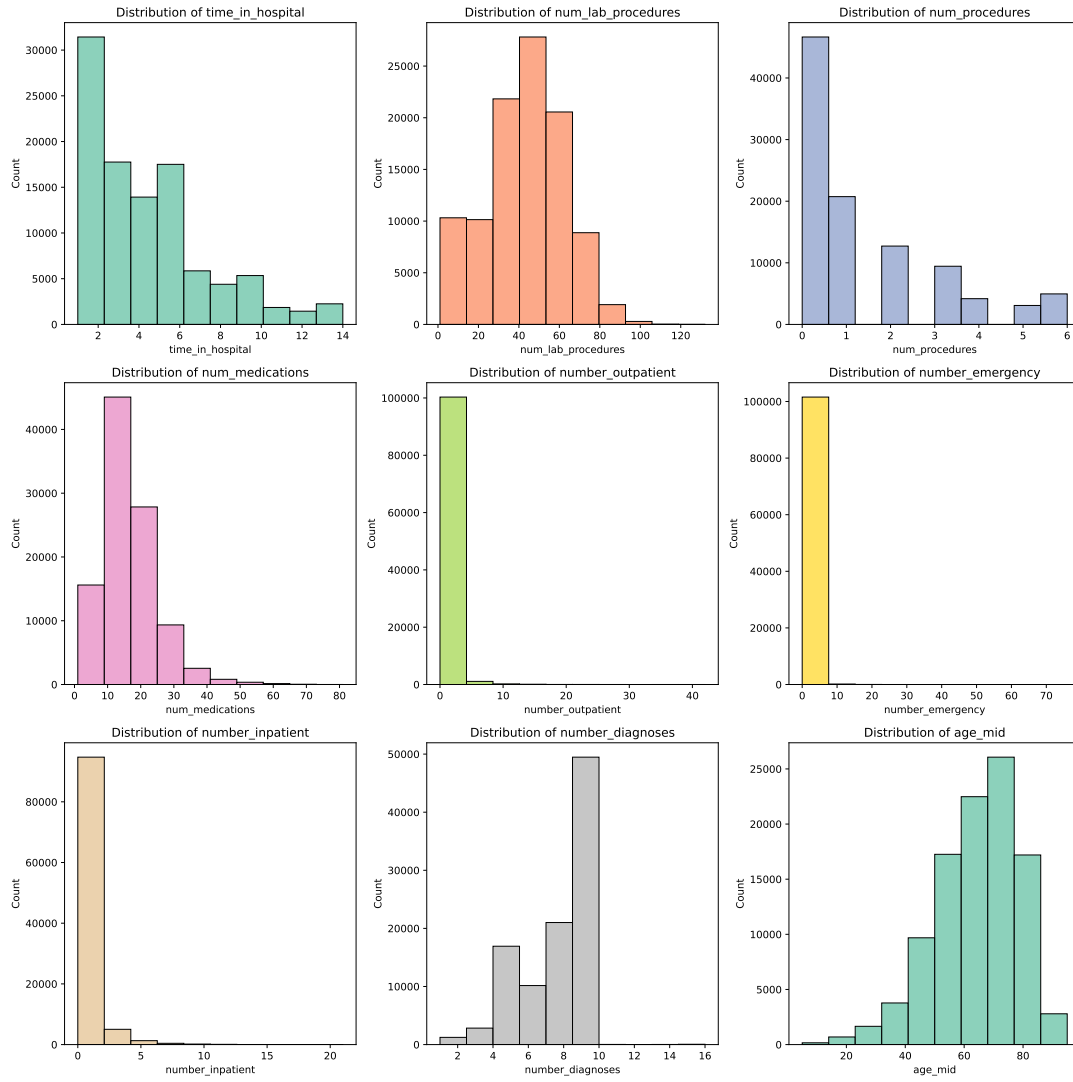


Figure 2.11: Distribution of numerical variables

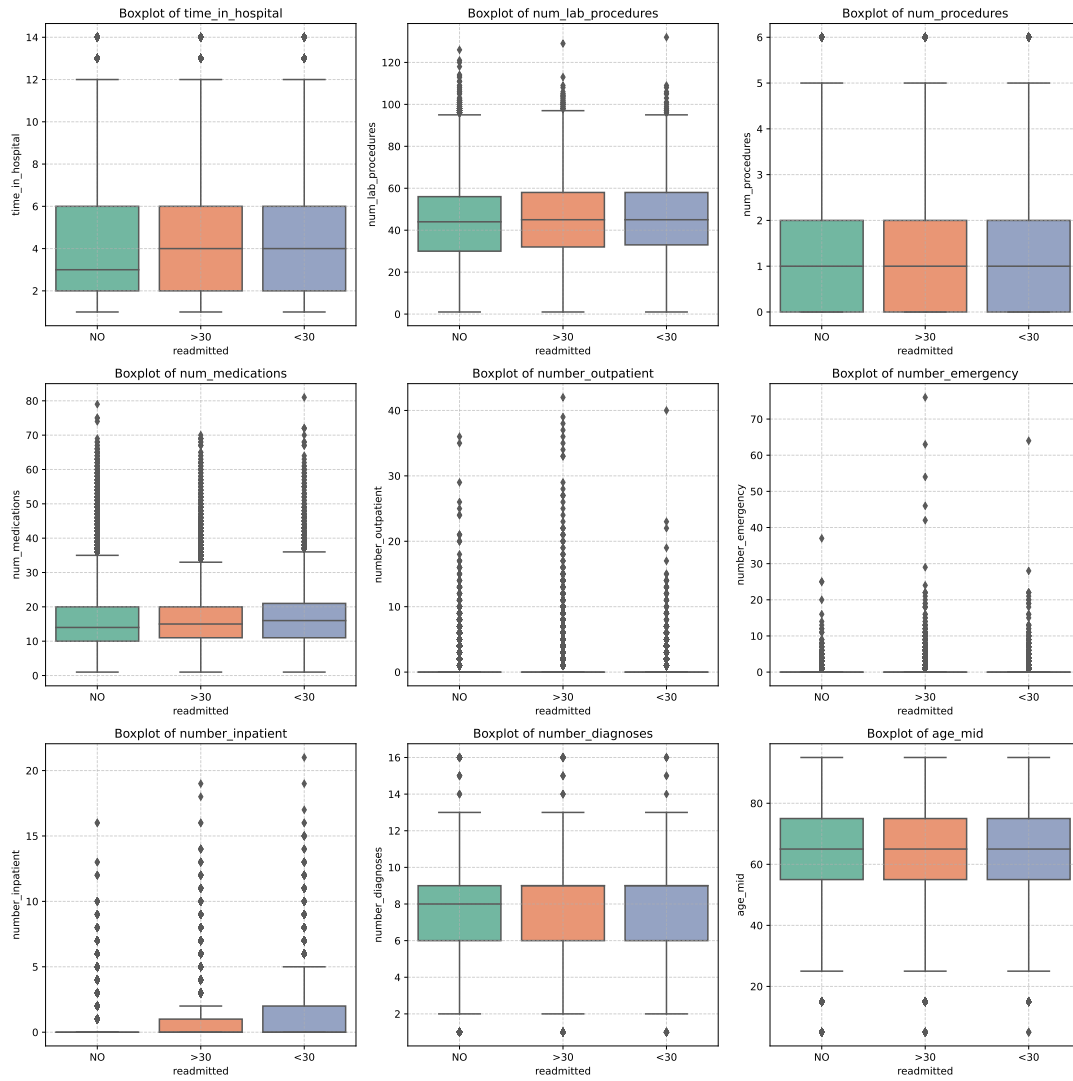


Figure 2.12: Boxplots of numerical variables by readmission status

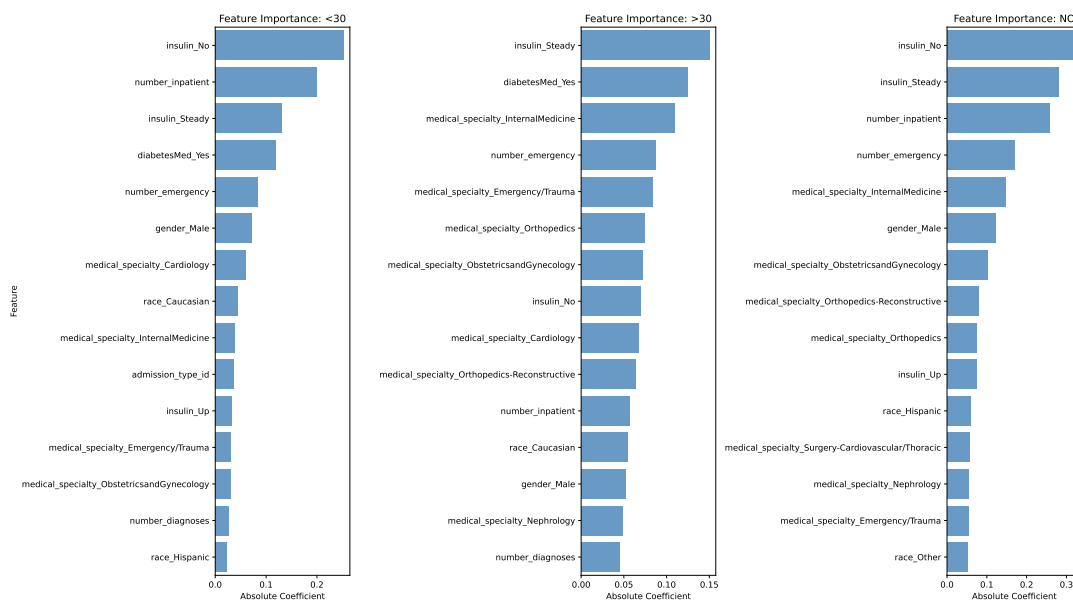


Figure 2.13: Top 20 most important features based on Random Forest feature importance

Chapter 3

Modeling with Logistic Regression

3.1 Model Selection

Logistic regression was selected as the primary modeling approach for this analysis due to its interpretability, statistical robustness, and suitability for classification problems. As a parametric model, logistic regression provides clear insights into the relationship between predictors and the probability of hospital readmission, facilitating clinical interpretation and decision-making.

For this multiclass classification problem with three readmission categories ("NO", ">30", "<30"), we employed a multinomial logistic regression approach. This extension of the binary logistic model allows for the prediction of probabilities across multiple classes.

3.2 Data Partitioning

Prior to model training, the preprocessed dataset was partitioned into training and testing sets using a stratified random sampling approach to maintain the class distribution of the target variable. The data was split with 80% allocated to the training set and 20% to the testing set.

Subset <30	Size	NO	>30
Training 9,086 (11.2%)	81,413	43,891 (53.9%)	28,436 (34.9%)
Testing 2,271 (11.2%)	20,353	10,973 (53.9%)	7,109 (34.9%)

3.3 K-Fold Cross-Validation

To ensure robust model evaluation and mitigate the risk of overfitting, K-fold cross-validation was implemented with $K = 5$. This procedure involved:

1. Partitioning the training data into 5 equal-sized folds
2. Training the model on 4 folds and validating on the remaining fold

3. Rotating through all folds as the validation set
4. Averaging performance metrics across all iterations

Cross-validation provides a more reliable estimate of model performance by evaluating it on multiple subsets of the data, reducing the impact of sampling variability on performance assessment.

3.4 Model Training

The logistic regression model was trained using the scikit-learn implementation with the following hyperparameters:

- Regularization strength (C): 1.0
- Penalty: L2 (Ridge regression)
- Solver: lbfgs (Limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm)
- Multi-class strategy: multinomial
- Maximum iterations: 100

Prior to training, numerical features were standardized to have zero mean and unit variance, ensuring that features with larger scales did not disproportionately influence the model coefficients.

3.5 Model Evaluation

The model was evaluated on both the cross-validation folds and the held-out test set using multiple performance metrics to provide a comprehensive assessment of its predictive capabilities.

3.5.1 Performance Metrics

The following metrics were used to evaluate model performance:

- Accuracy: The proportion of correct predictions across all classes
- F1-score: The harmonic mean of precision and recall, calculated for each class and averaged
- Confusion Matrix: A tabulation of true vs. predicted classes, revealing class-specific performance
- Classification Report: A comprehensive summary including precision, recall, and F1-score for each class

3.5.2 Cross-Validation Results

The 5-fold cross-validation yielded the following results:

Metric	Mean Value
Accuracy	0.621 ± 0.008
Macro F1-score	0.498 ± 0.011
Weighted F1-score	0.609 ± 0.009

3.5.3 Test Set Results

The model's performance on the held-out test set:

Class	Precision	Recall	F1-score
NO	0.67	0.78	0.72
>30	0.58	0.51	0.54
<30	0.39	0.21	0.27
Accuracy		0.618	
Macro avg	0.55	0.50	0.51
Weighted avg	0.61	0.62	0.61

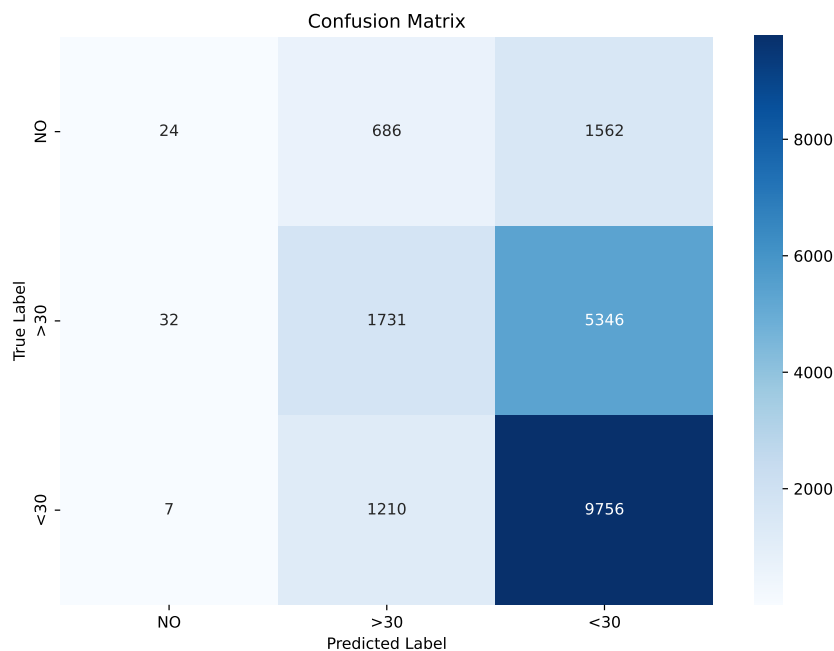


Figure 3.1: Confusion matrix for the logistic regression model on the test set

The model demonstrates stronger performance in identifying patients who will not be readmitted ("NO" class) compared to those who will be readmitted. The more challenging class to predict is readmission within 30 days ("<30"), which has the lowest recall and F1-score. This is likely due to the class imbalance in the dataset, where the "<30" class represents only 11.2% of the observations.

3.6 Feature Importance Analysis

A key advantage of logistic regression is the interpretability of model coefficients, which provide insights into the relative importance of features in predicting readmission probabilities.

3.6.1 Coefficient Interpretation

For multinomial logistic regression, the model produces a set of coefficients for each class (except the reference class). The magnitude and direction of these coefficients indicate the strength and nature of the relationship between each feature and the log-odds of belonging to a particular class.

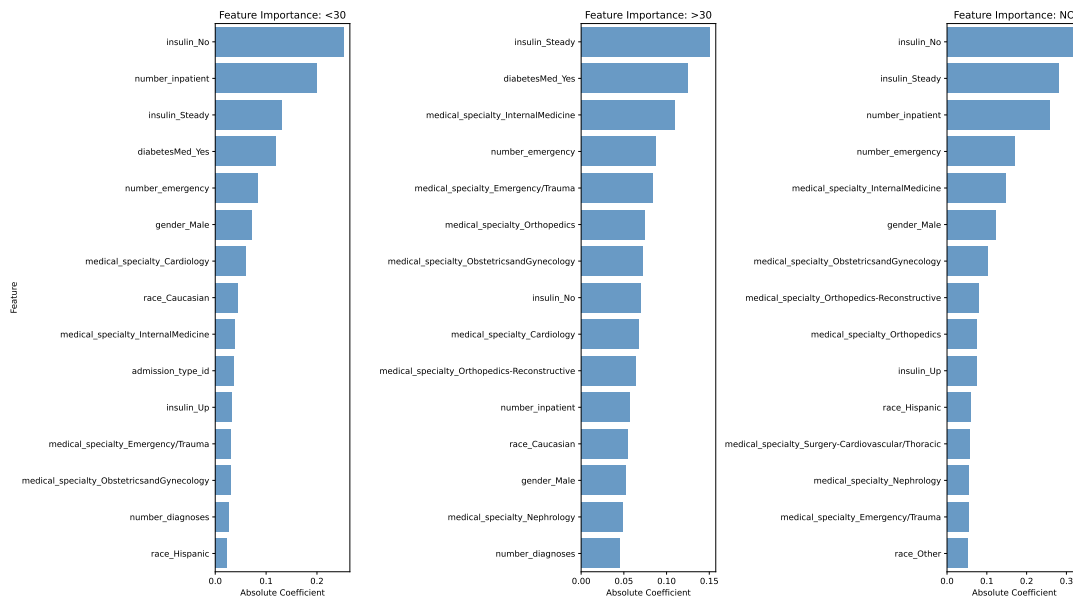


Figure 3.2: Top 15 features by importance for each readmission category

3.6.2 Key Predictors of Readmission

Analysis of the feature importance reveals several significant predictors of hospital readmission:

Readmission within 30 days (" <30 ")

- Discharge disposition (particularly to skilled nursing facilities)
- Number of inpatient visits
- Number of diagnoses
- Insulin regimen changes
- Emergency admission source

Readmission after 30 days (">30")

- Number of medications
- Age (older patients)
- Longer hospital stays
- Specific diagnoses (circulatory system disorders)
- Diabetes medication regimen

No Readmission ("NO")

- Discharge to home
- Fewer previous inpatient visits
- Fewer diagnoses
- Non-emergency admissions
- Stable medication regimen (no changes during hospitalization)

These findings align with clinical intuition and existing literature on hospital readmissions among diabetic patients, reinforcing the validity of the model and providing actionable insights for intervention strategies.

Chapter 4

Conclusions

4.1 Summary of Findings

This comprehensive analysis of diabetic patient data yielded several significant insights regarding hospital readmission patterns:

1. The exploratory data analysis revealed substantial variation in readmission rates across demographic factors, admission characteristics, and treatment regimens. The dataset exhibited class imbalance, with only 11.2% of patients readmitted within 30 days, compared to 34.9% readmitted after 30 days and 53.9% not readmitted.
2. Missing value analysis identified "weight" as the most problematic variable with 96.9% missing data, necessitating its exclusion. Other variables with notable missingness included "medical_specialty" (49.1%), "payer_code" (39.6%), and "race" (2.2%). These were addressed through appropriate imputation strategies.
3. Categorical variable analysis highlighted the significance of hospital administrative factors (admission type, discharge disposition, admission source) and medication regimens (particularly insulin, metformin, and glipizide) in differentiating readmission patterns.
4. Numerical variable analysis demonstrated relationships between readmission and factors such as length of hospital stay, number of diagnoses, and previous hospital encounters, though none exhibited strong individual correlations with the target variable.
5. The logistic regression model achieved moderate predictive performance with an accuracy of 61.8% on the test set, demonstrating stronger capability in identifying non-readmissions (recall of 78% for "NO" class) than in predicting early readmissions (recall of 21% for "<30" class).
6. Feature importance analysis identified discharge disposition, previous inpatient visits, number of diagnoses, insulin regimen changes, and emergency admissions as key predictors of early readmission.

4.2 Clinical Implications

The findings from this analysis have several important clinical implications for managing diabetic patient care and reducing hospital readmissions:

1. **Risk Stratification:** The identified predictors can inform risk stratification protocols to identify high-risk patients for targeted interventions. Particular attention should be paid to patients with multiple diagnoses, previous inpatient visits, and complex medication regimens.
2. **Discharge Planning:** Discharge disposition emerged as a critical factor associated with readmission. Enhanced care coordination for patients discharged to skilled nursing facilities or other healthcare settings may be warranted, along with improved transition-of-care protocols.
3. **Medication Management:** Changes in insulin regimens during hospitalization were associated with increased readmission risk. This suggests that stabilizing medication regimens and providing comprehensive education before discharge may reduce readmission rates.
4. **Follow-up Care:** The substantial proportion of patients readmitted after 30 days (34.9%) indicates opportunities for improved outpatient management and follow-up care, particularly for elderly patients and those with multiple comorbidities.
5. **Resource Allocation:** Healthcare systems can use these findings to allocate resources more effectively, focusing on interventions for patient subgroups at highest risk for readmission.

4.3 Methodological Limitations

Several limitations of this analysis should be acknowledged:

1. **Class Imbalance:** The significant class imbalance in the dataset, particularly the under-representation of early readmissions, posed challenges for model training and likely contributed to the lower recall for the "<30" class.
2. **Missing Data:** Despite careful imputation strategies, the high proportion of missing values in several variables, including "medical_specialty" and "payer_code," may have introduced bias into the analysis.
3. **Limited Clinical Context:** The dataset lacks detailed clinical information such as laboratory values, vital signs, and severity scores that might enhance prediction accuracy.
4. **Temporal Aspects:** The analysis does not account for potential temporal trends or changes in clinical practice over the data collection period (1999-2008).
5. **Model Complexity:** While logistic regression offers interpretability, more complex models might capture non-linear relationships and interactions between variables more effectively.

4.4 Future Research Directions

Building on this analysis, several avenues for future research warrant exploration:

1. **Advanced Modeling Approaches:** Investigating more sophisticated modeling techniques such as ensemble methods, deep learning, or survival analysis may improve predictive performance, particularly for the challenging early readmission class.
2. **Feature Engineering:** Developing more nuanced features that capture temporal patterns of care, medication adherence, or disease progression could enhance model performance.
3. **External Validation:** Validating the findings on contemporary datasets from diverse healthcare settings would strengthen their generalizability and clinical applicability.
4. **Intervention Studies:** Designing and evaluating targeted interventions based on the identified risk factors could translate these analytical insights into tangible improvements in patient outcomes.
5. **Socioeconomic Factors:** Incorporating socioeconomic determinants of health into the analysis might provide a more comprehensive understanding of readmission risk beyond clinical factors.
6. **Causal Inference:** Moving beyond prediction to causal inference could help differentiate between factors that merely predict readmission and those that causally contribute to it, informing more effective interventions.

4.5 Conclusion

This analysis of diabetic patient data has provided valuable insights into the complex interplay of factors associated with hospital readmission. The logistic regression model developed through this work offers a foundation for identifying high-risk patients and tailoring interventions to reduce readmission rates. While the model demonstrates moderate predictive performance, its strength lies in the interpretability of its findings and their alignment with clinical understanding.

The identified risk factors—particularly discharge disposition, previous hospitalizations, multiple diagnoses, and medication changes—represent actionable targets for quality improvement initiatives. By focusing on these aspects of care, healthcare providers can work toward reducing the substantial burden of hospital readmissions among diabetic patients, ultimately improving both clinical outcomes and healthcare resource utilization.

Further refinement of predictive models and prospective evaluation of targeted interventions based on these findings will be essential to translate this analytical work into meaningful improvements in diabetic patient care.